

STATISTICS 5.1.1. Representation of data

1. Types of data

Qualitative
(non-numerical)

- blood types:
A, B, O, AB
- colors



Quantitative
(numerical)

discrete

- e.g.:
- # of letters in a word:
1, 2, 4, 8, 14, ...
 - shoe size: $7\frac{1}{2}$, 9, 10, ...



continuous

- any value in the range
e.g. height (cm): 160.2, 160.24, ...



2. Stem - and - leaf diagrams

- small amounts of **discrete data**

- e.g.: Physics grade: 58, 55, 58, 61, 72, 79, 97, 67, 61, 77, 92, 64, 69, 62 and 53.

stem

- equal class widths
- ascending order

5	3 5 8 8	Key: 5 3 represents a score of 53%
6	1 1 2 4 7 9	
7	2 7 9	
8		
9	2 7	

leaf

- single digits only
- ascending order

EXAMPLE

The heights of boys and girls in a year 11 class are given to the nearest cm in the **back-to-back stem and leaf diagram** below. Write out the data in full.

First boy 8|16|, has height 168 cm. The boys are read backwards.

Key 8|16|5 means
Boys 168 cm and girls 165 cm

Boys

15
8 16
9, 8, 1 17
5, 2 18
1 19

Girls

15 9
1, 5, 9 16
0, 2, 3, 5 17
0 18
19

First girl, 15|9, has height 159 cm.

Boys: 168, 171, 178, 179, 182, 185, 191

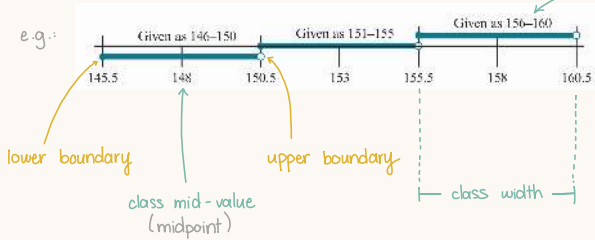
Girls: 159, 161, 165, 169, 170, 172, 173, 175, 180

3. Histograms

• continuous data * discrete data too

• numbers are rounded up/down to fit into classes

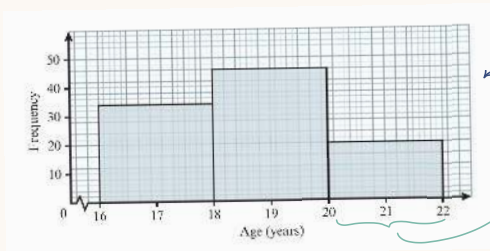
class



Ex 1.

Age (t years)	$16 \leq t < 18$	$18 \leq t < 20$	$20 \leq t < 22$
No. students (f)	34	46	20

grouped frequency table



frequency diagram

• no gaps between columns

= class width

"proportional to"

	First	Second	Third
Area	$2 \times 24 = 65$	$2 \times 46 = 92$	$2 \times 20 = 40$
Frequency	34	46	20

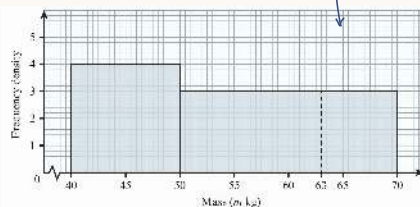
area $\propto$ frequency

Ex 2

Mass (m kg)	$40 \leq m < 50$	$50 \leq m < 70$
No. children (f)	40	60

Mass (m kg)	$40 \leq m < 50$	$50 \leq m < 70$
No. children (f)	40	60
Class width (kg)	10	20
Frequency density	$40 \div 10 = 4$	$60 \div 20 = 3$

histogram



$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

= frequency per standard interval

$$\text{frequency} = \text{f.density} \times \text{class width}$$

* assume values are evenly spread in whole class interval

4. Cumulative frequency graphs

continuous data

Ex 1.

boundaries:

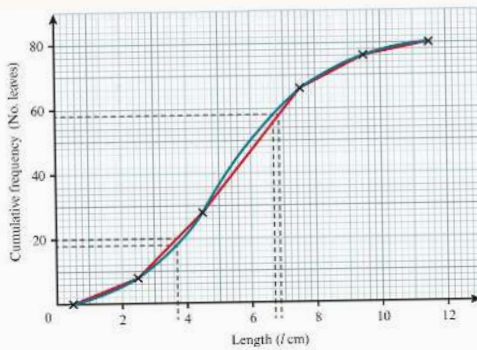
$0.5 \leq l < 2.5$

Lengths (cm)	1-2	3-4	5-7	8-9	10-11
No. leaves (f)	8	20	38	10	4

Lengths (l cm)	Addition of frequencies	No. leaves (cf)
$l < 0.5$	0	0
$l < 2.5$	$0 + 8$	8
$l < 4.5$	$0 + 8 + 20$	28
$l < 7.5$	$0 + 8 + 20 + 38$	66
$l < 9.5$	$0 + 8 + 20 + 38 + 10$	76
$l < 11.5$	$0 + 8 + 20 + 38 + 10 + 4$	80

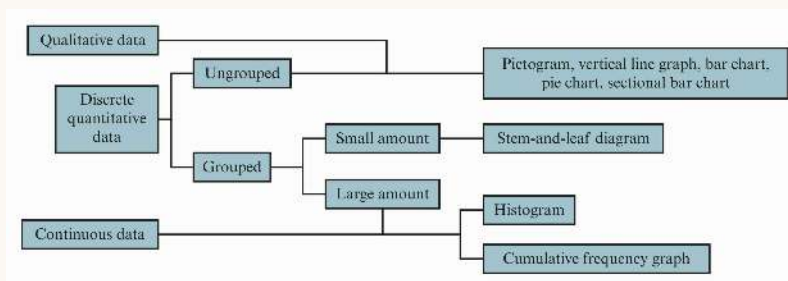
upper boundary

cf = cumulative frequency
= Σ frequency of all values up to a certain point



* draw a curved line to connect the dots

5. Comparing different data representations



STATISTICS 5.1.2 Measures of Central Tendency

1. Three types of averages

EXAMPLE Find the mean, median and mode of the following list of data: 2, 3, 6, 2, 5, 9, 3, 8, 7, 2

Put in order first: 2, 2, 2, 3, 3, 5, 6, 7, 8, 9

Mode = 2

mode :
most frequently occurring data value

Mean = $\frac{2+2+2+3+3+5+6+7+8+9}{10} = 4.7$

Median = average of 5th and 6th values = 4

$$\text{mean} = \bar{x} = \frac{\sum x}{n} = \frac{\sum fx}{\sum f}$$

sum of values
number of values

sum ("sigma") frequency

$$\text{median} = \left(\frac{n+1}{2} \right) \text{th value}$$

the middle value when all values are placed in order of size

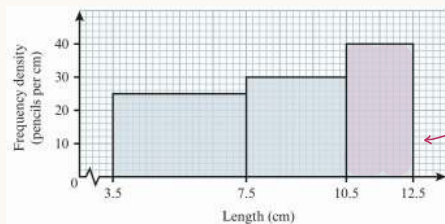
2. Mode and modal class

Ex: lengths of 270 pencils

Length (x cm)	No. pencils (f)
4-7	100
8-10	90
11-12	80



Length (x cm)	No. pencils (f)	Width (cm)	Frequency density
$3.5 \leq x < 7.5$	100	4	$100 \div 4 = 25$
$7.5 \leq x < 10.5$	90	3	$90 \div 3 = 30$
$10.5 \leq x < 12.5$	80	2	$80 \div 2 = 40$



modal class: class with highest frequency density

3. Means from grouped frequency tables

→ you'll have to estimate using mid-class values

Height of tree to nearest m	Mid-class value x	Number of trees f	fx
0-5	2.75	26 (26)	71.5
6-10	8	17 (43)	136
11-15	13	11	143
16-20	18	6	108
	Totals	60 ($= \sum f$)	458.5 ($= \sum fx$)

Lower class boundary = 0.
Upper class boundary = 5.5.
So the mid-class value = $(0 + 5.5) \div 2 = 2.75$.

$$\text{Estimated mean} = \frac{458.5}{60} = 7.64 \text{ m}$$

* **coded data** : transform values (adding a +ve/-ve constant)
 ↳ **easier to handle**

Ex: Find mean of 101, 103, 104, 109, 113

$$\begin{array}{c} \downarrow -100 \\ \textcircled{1, 3, 4, 9, 13} \\ \downarrow \bar{x} = \frac{1+3+4+9+13}{5} + 100 = 6 + 100 = 106 \end{array}$$
 put it back!

For **ungrouped data**,

$$\bar{x} = \frac{\sum(x-b)}{n} + b$$

For **grouped data**,

$$\bar{x} = \frac{\sum(x-b)f}{\sum f} + b$$

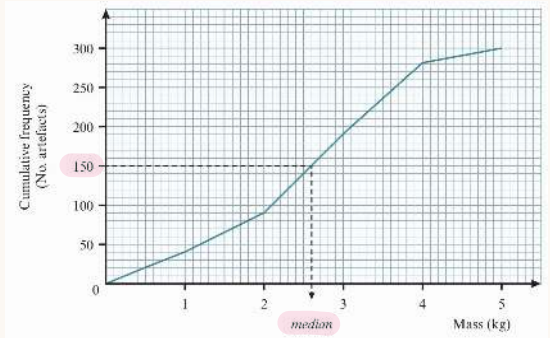
$$\bar{x} = \text{mean}(x-b) + b$$

4. **Median for grouped data** → we estimate, yet again, with a **graph**

Ex.: masses of 300 museum artefacts

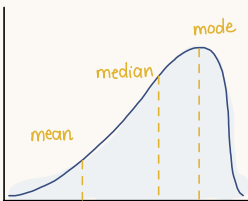
for an estimate,

$$\text{median} = \frac{n}{2} \text{th value}$$



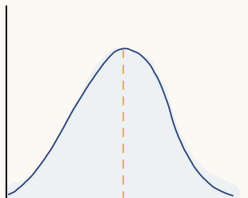
* **skewness** (which average is heavily affected by outliers?)

negative skew



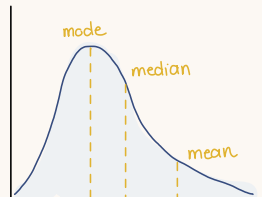
$$\text{mean} < \text{median} < \text{mode}$$

symmetrical



$$\text{mean} = \text{median} = \text{mode}$$

positive skew



$$\text{mode} < \text{median} < \text{mean}$$

Measures of Variation

↳ to measure: how widely spread out the values are. / dispersion

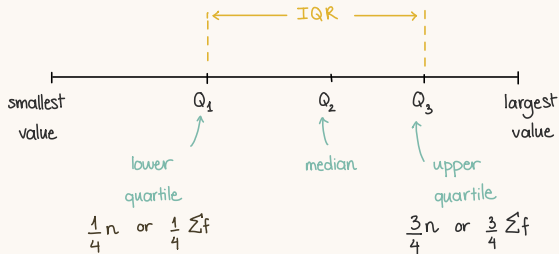
1. Range range = highest value - lowest value

* heavily affected by extremes / outliers

2. Interquartile range + percentiles (IQR)

$$IQR = Q_3 - Q_1$$

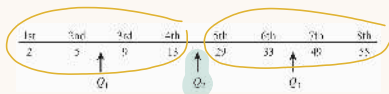
* relatively unaffected by outliers



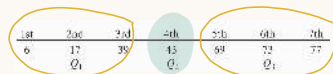
* for grouped data
total frequency $n = \sum f$

* for ungrouped data

- even # of values

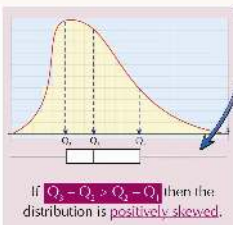
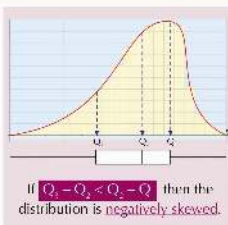
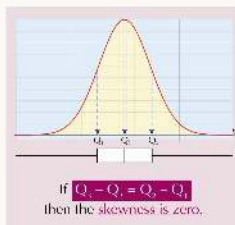
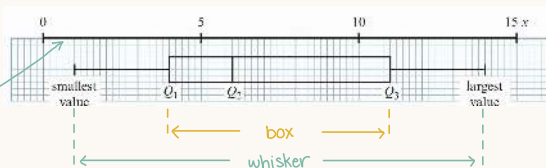


- odd # of values



* box - and - whisker diagrams

scale



Var

SD

3. Variance and standard deviation

↳ measures: how close a value is to the mean

$$\text{standard deviation} = \sqrt{\text{variance}}$$

• for ungrouped data :

$$\bar{x} = \frac{\sum x}{n} ;$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2}$$

• for grouped data :

$$\bar{x} = \frac{\sum fx}{\sum f} ;$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2 f}{\sum f}} = \sqrt{\frac{\sum x^2 f}{\sum f} - \bar{x}^2}$$

EXAMPLE Find the mean and standard deviation of the following numbers: 2, 3, 4, 4, 6, 11, 12

1) Find the **total** of the numbers first: $\sum x = 2 + 3 + 4 + 4 + 6 + 11 + 12 = 42$

2) Then the **mean** is easy: Mean $= \bar{x} = \frac{\sum x}{n} = \frac{42}{7} = 6$

3) Next find the **sum of the squares**: $\sum x^2 = 4 + 9 + 16 + 16 + 36 + 121 + 144 = 346$

4) Use this to find the **variance**: Variance, $s^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{346}{7} - 6^2 = 49.43 - 36 = 13.43$

5) And take the **square root** to find the standard deviation: Standard deviation $= \sqrt{13.43} = 3.66$ to 3 sig. fig.

✕ **combined sets of data**

$$\text{mean of } x \text{ and } y \text{ combined} = \frac{\sum x + \sum y}{n_x + n_y}$$

$$\text{variance of } x \text{ and } y \text{ combined} = \frac{\sum x^2 + \sum y^2}{n_x + n_y} - \left(\frac{\sum x + \sum y}{n_x + n_y} \right)^2$$